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Review

Effects of mindfulness-based interventions (MBIs) on psychological outcomes and HbA1C in patients with diabetes: A systematic review and meta-analysis of randomized controlled trials

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ABSTRACT

We synthesized the effects of MBIs on psychological outcomes and HbA1c in diabetic patients and conducted subgroup analyses to investigate the sources of heterogeneity. Nine electronic databases were methodically explored from their inception through May 2025. We reviewed studies on the outcomes for individuals with diabetes mellitus who participated in mindfulness-based interventions. A random-effects model was employed to calculate the effect size. The funnel plot, Q statistics, and I2 were employed to assess heterogeneity among studies. Across 24 primary studies ($N = 1987$), 942 participants engaged in mindfulness-based interventions and 955 functioning as controls. Mean age ranged from 18.3 to 66.3 years. Overall, mindfulness-based interventions showed improvement of HbA1c (Hedges' $g = 0.564$, 95%Confidence interval, CI, 0.221, 0.908, $p = 0.001$), stress ($g = 1.132$, 95%CI 0.551, 1.714, $p < 0.001$), anxiety ($g = 1.321$, 95%CI 0.569, 2.072, $p = 0.001$) and depression ($g = 1.059$, 95%CI 0.564, 1.555, $p < 0.001$). MBI format, intention-to-treat analysis, concealed allocation, attrition, funding sources, and a priori analysis were modifiers affecting the effect size. To sum up, mindfulness-based therapies markedly improved HbA1c levels and psychological outcomes, among individuals with diabetes mellitus. Future studies with robust designs are warranted. Healthcare professionals may consider using these therapies to improve psychological outcomes and HbA1c levels in individuals with diabetes mellitus.

1. Introduction

Diabetes mellitus, including both type 1 and type 2, is a global health challenge marked by persistent hyperglycemia, which can lead to various microvascular and macrovascular complications [1,2]. Elevated glycosylated hemoglobin (HbA_1c) is a key indicator of poor glycemic control and is closely associated with increased risks of complications, morbidity, and mortality [3–6]. Additionally, individuals with diabetes often experience psychological distress, which includes elevated levels of depression, anxiety, and diabetes-specific distress. These emotional challenges can hinder effective disease management and negatively impact overall quality of life [7–9].

Traditional therapeutic approaches for managing diabetes typically focus on pharmacologic glycemic control and lifestyle modifications. However, these traditional therapeutic approaches often neglect the psychological aspects of diabetes [10]. A variety of non-mindfulness-based psychosocial interventions have been developed to support psychological well-being among individuals with T1D and T2D [11–13]. Cognitive-behavioral therapy (CBT) has demonstrated effectiveness in reducing depressive symptoms and improving glycemic outcomes [14]. Diabetes self-management education and support (DSMES) programs aim to enhance knowledge, self-efficacy, and coping skills, leading to improvements in both psychological and clinical outcomes [15,16]. In addition, social work interventions and peer support programs, including group-based and community-based models, have been shown to reduce social isolation, enhance emotional support, and improve diabetes-related outcomes [17,18]. Despite these benefits, such interventions may be limited by issues related to accessibility, resource intensity, and long-term sustainability, particularly in rural or low-resource settings [14,19,20].

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In recent years, mindfulness-based interventions (MBIs) have emerged as a promising approach for addressing both psychological and physiological aspects of chronic disease management [21–23]. Mindfulness-based interventions (MBIs), such as mindfulness-based stress reduction (MBSR) and mindfulness-based cognitive therapy (MBCT), present promising complementary strategies that address both metabolic and psychological health [22,24]. MBIs promote nonjudgmental awareness of present-moment experiences, emotional regulation, and self-compassion, which may lead to improved self-care and enhanced stress resilience [21,25,26].

Recent high-quality evidence encapsulates the emerging benefits of MBIs in diabetes care. A comprehensive systematic review and *meta*-analysis conducted in 2025, which included 31 randomized controlled trials (RCTs) with a total of 2,337 participants, concluded that MBIs may modestly reduce HbA_1c levels, with a mean difference of −0.44 (95% CI: −0.71 to −0.17; sample size = 9; low certainty). More substantial effects were observed in psychological outcomes, including stress (SMD = -1.01, 95% CI: −1.91 to −0.20; sample size = 8), depression (SMD = -1.26, 95% CI: −2.08 to −0.43; sample size = 7), and anxiety (SMD = -0.67, 95% CI: −1.27 to −0.08; sample size = 4), compared to usual care or waitlist controls [21]. However, this *meta*-analysis had a limited number of primary studies (sample sizes ranging from 4 to 9), which could lead to overestimated effect sizes. The small number of included studies also poses a limitation for conducting subgroup analyses to explore the sources of heterogeneity across the studies.

A *meta*-analysis that included seven randomized controlled trials (RCTs) with a total of 987 participants found a significant reduction in HbA_1c levels, with a mean difference of –0.50% (95% CI –0.67 to –0.34) at 12 to 20 weeks post-intervention. Additionally, there was a notable reduction in depression, indicated by a standard mean difference (SMD) of –0.84 (95% CI –1.11 to –0.56)[22]. Similarly, a 2020 review of mindfulness and acceptance-based therapies for individuals with type 2 diabetes, which included nine RCTs with 801 participants, reported reductions in diabetes-related distress (SMD –0.37; 95% CI –0.63 to –0.12) and HbA_1c levels (mean difference –0.35%; 95% CI –0.67 to –0.04) up to one month after the intervention [27]. A comprehensive review of systematic studies revealed significant psychological benefits of MBIs. These benefits included notable reductions in depression (Standardized Mean Difference [SMD] of −0.84; 95% Confidence Interval [CI] of −1.16 to −0.51), stress (SMD of −0.53; 95% CI of −0.75 to −0.31), and diabetes-related distress (with a mean difference of approximately −5.8). Additionally, improved mental quality of life was observed. However, the effects of MBIs on anxiety and self-management were less consistent [28].

These findings indicate that while MBIs show potential for modest reductions in HbA1c and thus offer glycemic benefits, their most significant effect seems to be in alleviating the psychological burden experienced by individuals with diabetes. However, current studies differ regarding the type of intervention, dosage, follow-up duration, and types of diabetes examined. Additionally, many studies are limited by small sample sizes or inconsistent outcomes.

Therefore, this systematic review and *meta*-analysis aim to thoroughly evaluate the effectiveness of mindfulness-based interventions (MBIs) on two key outcomes for patients with diabetes mellitus:

1. Metabolic control, as indicated by changes in HbA_1c levels, and
2. Psychological outcomes, including stress, anxiety, and depression.

By applying rigorous inclusion criteria, we seek to combine evidence from recent randomized controlled trials (RCTs) and quantify the magnitude and certainty of the effects. Our goal is to inform clinical practice and guide future research directions.

2. Methods

This study was guided by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) methodology, which enabled the identification, selection, and critical appraisal of the literature [29,30]. A study protocol was registered with the International Prospective Register of Systematic Reviews, PROSPERO (CRDxxxxxxxxx).

2.1. Search strategy and selection criteria

Data from nine databases, namely Academic Search Complete (1990 +), CINAHL (1937 +), Cochrane (1995 +), APA PsycINFO (1967 +), Ovid Medline (1946 +), ProQuest & Theses (1996 +), PubMed (1809 +), Web of Science (1880 +), and Scopus (1788 +), was analyzed from their inception through May 2025. Additionally, we analyzed the reference lists of eligible studies (ancestry search). The following search terms employed were: (mindfulness OR mindful* OR meditat* OR meditation) AND (diabetes OR (diabetes mellitus) OR diabetic patient*). Employing truncation with an asterisk contained all variations of the phrases and enabled the retrieval of a greater number of articles having analogous terminology. Furthermore, subject headings were analyzed to obtain a more comprehensive set of results.

2.2. Inclusion and exclusion criteria

We employed the PICOS framework—population, intervention, comparison, outcome, and study design—in our systematic review and *meta*-analysis [31]. Only articles published in English were selected for this study.

The population selected for review included general adults diagnosed with diabetes mellitus.

Intervention: Mindfulness-based therapies rooted in Buddhist principles, which primarily involved formal meditation practices such as Mindfulness-Based Stress Reduction (MBSR), Mindfulness-Based Cognitive Therapy (MBCT), and modified Mindfulness-Based Interventions (MBIs).

Comparison: Standard treatment, waitlist control, and control groups were used for comparison.

Outcome: The outcomes measured included quantitative levels of HbA1c, as well as levels of stress, anxiety, and depression.

We omitted main studies that 1) incorporated MBIs alongside an additional intervention that could influence HbA1c and psychological outcomes (e.g., qigong, tai chi, yoga, psychotherapy, physical activity); and 2) were insufficient in complete data necessary for calculating effect sizes.

Two independent researchers, CR and SP, reviewed the titles and abstracts of the identified citations. They then searched for full-text papers that appeared potentially eligible according to either reviewer. Both researchers evaluated these full-text papers separately based on the established qualifying criteria. Any discrepancies in the selection of studies were resolved through a consensus discussion among the research team (CR, SP, and SO).

2.3. Data extraction and coding

After assessing all 24 primary studies, two researchers (CR and SP) developed a codebook and performed pilot testing on three of the studies. The data from each primary study was individually extracted and coded by each researcher. The data included the information source (e.g., country, year of publication, and publication status); methodologies included the study's quality assessment (e.g., blinded collector, intention-to-treat analysis, sample power, intervention fidelity, and baseline participant characteristics); intervention details (e.g., type of MBIs, delivery format, number of sessions per course, duration of each session in minutes, and components of the interventions); participant characteristics (e.g., mean age, gender distribution, sample size for each group, and health status); and results (e.g., HbA1c levels, mean and standard deviation of psychological outcomes).

2.4. Statistical analysis

We initially performed descriptive statistics using SPSS software to examine the research characteristics. We utilized Comprehensive Meta-Analysis software (CMA, version 4.0) to calculate standardized mean differences between the post-test outcomes of MBI groups and control groups, accompanied by 95% confidence intervals for effect size estimation [32]. This methodology was chosen because the diverse instruments employed provide outcomes related to HbA1C, stress, anxiety, and depression.

Considering the expected variability among research and the premise that our selected studies constituted a random sample from a normally distributed population of effect sizes, we employed a random-effects model [32]. In this model, CMA allocates weights to individual studies according to the inverse of their combined within-study and between-study variances, so ensuring proper weighting of each study's contribution to the pooled effect estimate [32].

We computed the pre/post standardized mean difference in psychological outcomes for a single group between the MBI and control groups to examine the possibility of spontaneous psychological lowering. If the control group had a reduction in psychological outcomes similar to that of the MBI group, we would conclude that MBI is no more effective than the control group or spontaneous recovery.

2.5. Heterogeneity assessment

The forest plot, Q statistic, and I^2 statistic were employed to investigate the heterogeneity among research [32]. The picture illustrates study heterogeneity as shown in the forest plot. The existence of heterogeneity is signified by a substantial Q statistic score. The I^2 criteria of 25%, 50%, and 75% signify low, moderate, and high heterogeneity, respectively. The most significant observed variation is the true variance when the I^2 value approaches 100% [32].

In the presence of heterogeneity, we conducted a subgroup analysis to explore its origins based on participant characteristics, quality measures, interventions, and information sources [32]. We utilized *meta*-regression, analogous to regression analysis for continuous moderators, and *meta*-analytic analysis of variance, analogous to analysis of variance for categorical moderators [32]. In instances where less than six primary studies existed within a subgroup, we computed a confidence interval for the average impact utilizing the Hartung, Knapp, Sidik, and Johkman (HKSJ) adjustment [33,34]. In cases of limited primary studies within subgroup analysis, the HKSJ adjustment yields a confidence interval that is broader and more precise. This method yields a more precise estimation of the average effect size than the traditional approach.

2.6. Assessment of methodological quality

We employed the Revised Cochrane 'Risk of Bias' assessment for randomized trials (RoB 2.0) to evaluate methodological quality [35]. Two researchers conducted a critical evaluation of the five probable biases. All potential biases in primary studies were classified as “high risk,” “low risk,” or “some concerns.” Furthermore, two researchers independently assessed quality indicators (e.g., concealed allocation, intention-to-treat analysis, sample size, baseline participant characteristics, and intervention fidelity) to evaluate the methodological quality of each primary study. Subsequently, to assess their influence on ESs, we analyzed them as moderators.

2.7. Quality of evidence assessment

The GRADE methodology is employed to designate a high, moderate, low, or very low grading to the overall quality of evidence in the *meta*-analysis [36,37]. The GRADE rating signifies the overall confidence in the effectiveness of interventions, transcending the risk of bias that pertains to the internal validity of the original studies. For randomized controlled trials, GRADE utilizes a high initial grade [37]. Five assessment domains may be employed to reduce this rating. The three researchers (CR, SP, and SO) assessed and deliberated the ratings.

2.8. Risk of publication bias

Egger's bias values, the Begg and Mazumdar rank correlation test, and a funnel plot were utilized to determine publication bias. A distribution exhibiting a visually symmetrical funnel shape indicates the absence of publishing bias. The rank order correlation (Kendall's τ) between variance and the standard treatment effect is determined using the Begg and Mazumdar test. Publication bias exists when the Begg and Mazumdar rank correlation test and Begg's bias yield significant values [32].

2.9. E**thical approval

As no original raw data was utilized and the data for the *meta*-analysis were sourced from primary studies that obtained ethical approval, neither ethical consent nor patient agreement is required for this *meta*-analysis.

3. Results

3.1. Demographic of studies

After the exclusion of duplicates, an initial pool of 1,408 articles from nine electronic databases was established. Out of 1422 publications, 1,357 were removed following title and abstract evaluation, leaving 65 articles for further review. Forty-one primary studies were excluded following the full-text evaluation, leaving twenty-four primary studies for this systematic review and *meta*-analysis. Refer to Fig. 1.

Twenty-four main studies qualified the inclusion criteria and were incorporated into this analysis, comprising data from 18 between-group comparisons for HbA1c, 17 for stress outcomes, 14 for depression outcomes, and 8 for anxiety outcomes. All studies were published between 2012 and 2025. Our systematic review and *meta*-analysis encompassed 1987 people, comprising 942 persons engaged in mindfulness-based interventions (MBIs) and 955 functioning as controls. Five primary studies took place in Iran [38–42], four in China [43–46], three in Netherlands [47–49], two each in the United States [50,51], Germany [52,53], and India [54,55], and one each in New Zealand [56], UAE [57], South Korea [58], Canada [59], Australia [60], and Thailand [61]. See supplementary table 2. The mean age ranges from 18.3 to 66.3 years (Mean = 51.85, SD = 13.55, s = 21. See Table 1.

MBIs primarily encompassed MBSR, MBCT, and modified MBIs. Refer to Table 1 for a description of the MBI, which includes the total weeks of intervention, the number of sessions per week, and the duration of each MBI session in minutes. Additionally, refer to supplementary Table 2.

3.2. Effects of MBIs

3.2.1. HbA1C

The overall effect size from 18 comparisons was 0.564 (95% CI 0.221, 0.908, $p = 0.001$, $I^2 = 89.70\%$), suggesting that MBIs had a moderate impact on improving HbA1c levels in individuals with diabetes mellitus in comparison with controls. Out of 18 comparisons, eight had a substantial positive effect improvement.

3.2.2. Stress

The overall effect size from 17 comparisons was 1.132 (95% CI 0.551, 1.714, $p < 0.001$, $I^2 = 95.88\%$), suggesting that MBIs significantly alleviated stress symptoms in patients with diabetes mellitus compared to controls. Out of 17 comparisons, nine had a substantial positive effect improvement.

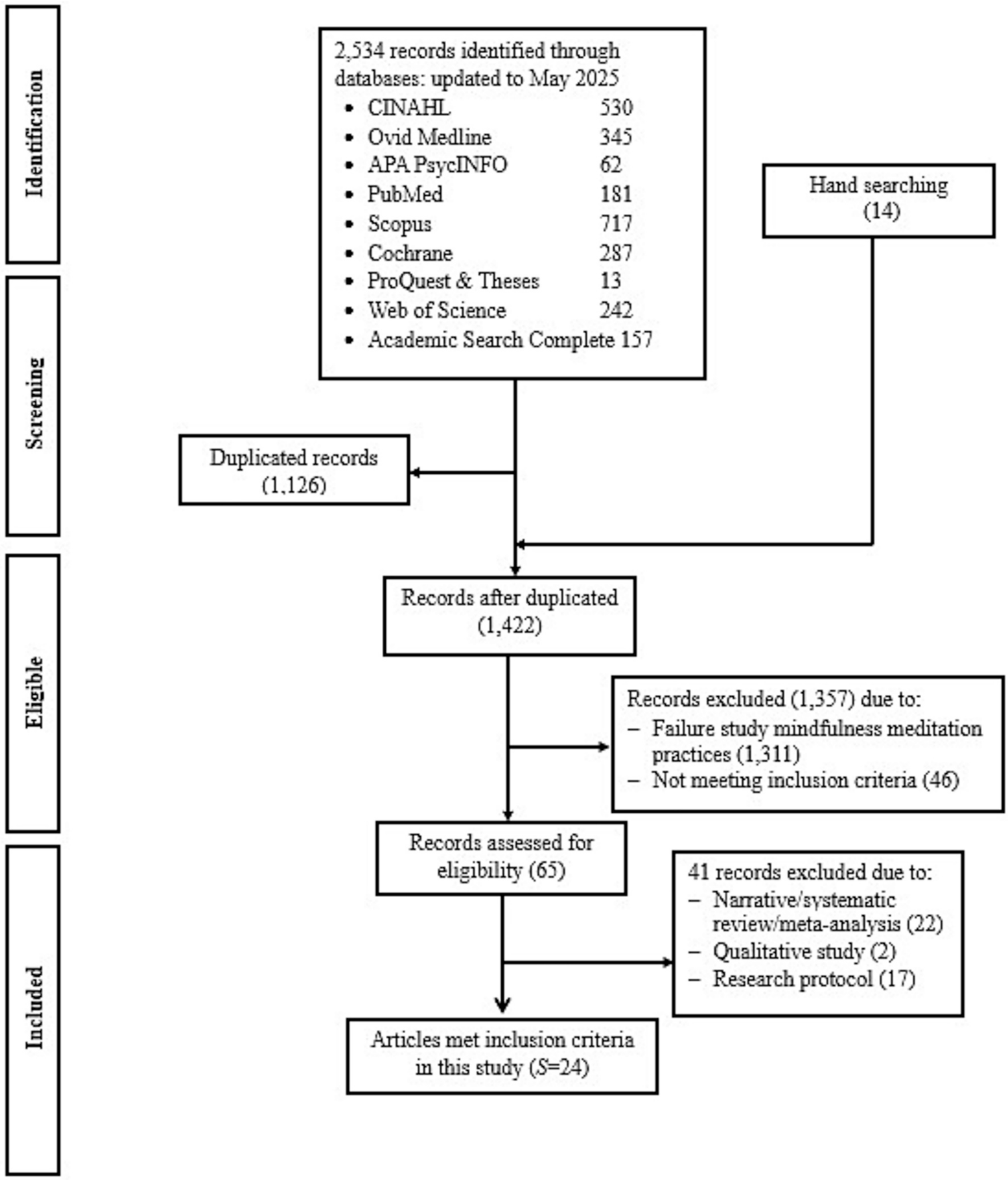

Fig. 1. PRISMA Flow of Included Primary Studies.

3.2.3. *Anxiety*

The overall effect size across eight comparisons was 1.321 (95% CI 0.569, 2.072, $p = 0.001$, $I^2 = 95.09\%$), signifying that MBIs significantly alleviated anxiety symptoms in individuals with diabetes mellitus compared to controls. Out of the eight comparisons, five exhibited significant positive enhancement.

3.2.4. *Depression*

The overall effect size across 14 comparisons was 1.059 (95% CI 0.564, 1.555, $p < 0.001$, $I2 = 93.64\%$), signifying that MBIs significantly improved depressive symptoms in individuals with diabetes mellitus compared to controls. Nine out of the fourteen comparisons exhibited significant positive improvement.

3.3. *Moderator analyses*

Because the heterogeneity across studies existed in all psychological outcomes and HbA1c, we did subgroup analyses to explore the source of heterogeneity.

For HbA1c, there were four moderators that influenced the effect size. Conducting MBIs with funding had a less effect size ($g = 0.259$) to improve HbA1c than conducting those without funding ($g = 1.219$, $p = 0.004$). Using intention-to-treat technique for analysis had a greater effect ($g = 0.922$) on improving HbA1c than non-using this technique in patients with diabetes mellitus ($g = 0.134$, $p = 0.019$). using concealed allocation techniques had a less effect ($g = 0.002$) to improve HbA1c in patients with diabetes mellitus than without technique ($g = 1.015$, $p = 0.001$). Interestingly, every one percent of attrition rate increased, the HbA1c level decreased by 0.063 U of effect size.

Table 1
Characteristics of Primary Studies ($s = 24$).

Characteristics	s	Min	Q1	Mdn.	Q3	Max	Mean	SD
Mean age (years)	21	18.3	44.9	56.8	61.6	66.3	51.85	13.55
Total Sample size at analysis								
–MBI group	24	12.0	23.25	42.0	50.0	70.0	39.25	17.74
Control group	24	13.0	25.50	41.0	50.75	69.0	39.79	17.78
Weeks of structured MBI	23	8	8	8	9	24	9.26	3.44
Days across intervention (length)	23	49	49.0	49.0	56.0	183.0	58.78	28.42
Structured MBI session/week	21	1.0	1.0	1.0	1.0	2.0	1.05	2.61
Structured MBI min./session	17	20.0	73.7	120.0	120.0	150.0	98.08	39.40
Dose (length x duration)	17	1470.0	4035.0	5880.0	6545.0	9240.0	5429.26	2010.36
Days after intervention measured	24	0	0	28.0	91.50	1095.0	115.83	260.91
% Attrition, MBI group	24	0	0	0	10.76	36.84	6.41	10.34
% Attrition, Control group	24	0	0	0	9.22	39.29	6.02	10.68

s = number of studies providing data, Min = minimum, Q1 = first quartile, Mdn = median, Q3 = third quartile, Max = maximum, MBI = mindfulness-based intervention

Table 2
Effect size of MBIs on HbA1c level and psychological outcomes compared with control groups.

Comparison	MBI group							
	k	ES	p(ES)	95% CI	SE	I^2	Q	p(Q)
MBI vs. Control groups								
HbA1c	18	0.564	0.001	0.221, 0.908	0.175	89.70	165.10	0<.001
Stress	17	1.132	0<.001	0.551, 1.714	0.297	95.88	387.90	0<.001
Anxiety	8	1.321	0.001	0.569, 2.072	0.384	95.09	142.41	0<.001
Depression	14	1.059	0<.001	0.564, 1.555	0.253	93.64	204.49	0<.001

For stress outcome, only one moderator influencing the effect size of MBIs in patients with diabetes mellitus. Using intention-to-treat technique had a greater effect ($g = 1.624$) to improve stress level in patients with diabetes mellitus than those without technique ($g = 0.245$, p.025).

For anxiety outcome, only one moderator influencing the effect size of MBIs in patients with diabetes mellitus. Providing MBIs as individual format had a greater effect ($g = 2.265$) to improve anxiety symptoms in patients with diabetes mellitus than providing MBIs as a group ($g = 0.770$, $p = 0.045$).

For depression outcome, there were three moderators influencing the effect size of MBIs in patients with diabetes mellitus. Using intention-to-treat technique had a greater effect ($g = 1.504$) to improve depressive symptoms than non-used technique ($g = 0.264$, $p = 009$). Also, using concealed allocation techniques had less effect ($g = 0.544$) to improve depression than non-used this technique ($g = 1.781$, $p = 0.015$). Lastly, using a priori analysis for recruiting participants had less effect ($g = 0.495$) to improve depressive symptoms than non-used technique ($g = 1.643$, $p = 0.020$).

4. Quality of methodological study

After we assessed the quality of methodological studies, all researchers conducted the trials with individual randomization. Six to eleven research teams reported that they used intention-to-treat techniques for data analysis ($s = 6–11$). Few researchers used high quality methodological indicators such as power of sample, and fidelity intervention. For instance, three to seven of 24 research teams reported that they had the power of sample for recruiting participants ($s = 3–7$). Only three to five of 24 research teams reported about the fidelity of intervention ($s = 3–5$).

Moreover, methodological quality risk of bias was evaluated for every primary study. Twenty-one of 24 primary studies were deemed to have some concerns, and only three to have a low risk of bias.

4.1. Quality of evidence assessment

The overall quality of evidence, using GRADE, indicated a moderate degree of confidence in the pooled effect estimates [62]. The overall quality assessment of the evidence was moderate, signifying a moderate level of confidence in the estimates of the pooled effects. The high degree of confidence among RCTs was diminished to a moderate level due to substantial concerns over publication bias, considerable heterogeneity, and inconsistency [63]. Overall, the possibility of bias, indirectness, and imprecision was not judged to be a significant problem.

4.2. Publication bias

For HbA1c outcome, the funnel plot for HbA1c outcomes exhibited visual symmetry, indicating a low likelihood of publication bias. Egger's test yielded a non-significant result with an intercept of 3.07 (95% CI −2.76, 8.90, $p = 0.140$). The Begg and Mazumdar test yielded a non-significant result ($p = 0.120$). These findings suggest an absence of publication bias.

For stress outcome, the funnel plot exhibited visual asymmetry, indicating a probable presence of publishing bias. Egger’s test yielded a significant intercept of 12.21 (95% CI 5.06, 19.35, $p = 0.001$). The Begg and Mazumdar test yielded a significant result ($p = 0.001$). These data suggest a possible existence of publication bias.

For anxiety outcome, the funnel plot exhibited visual symmetry, indicating that publication bias was improbable. Egger's test yielded a non-significant result with an intercept of 12.94 (95% CI −3.83, 29.71, $p = 0.054$). The Begg and Mazumdar test yielded a non-significant result ($p = 0.132$). These findings suggest the absence of publication bias.

For depression outcome, the funnel plot exhibited visual asymmetry, indicating a probable presence of publishing bias. Egger's test yielded a significant intercept of 7.49 (95% CI −0.15, 15.14, $p = 0.027$). The Begg and Mazumdar test yielded a significant result ($p = 0.069$). The data suggest a possible existence of publication bias.

5. Discussion

5.1. Effects of MBIs

To our knowledge, this is the first systematic review and *meta*-analysis to comprehensively estimate the effectiveness of MBIs on HbA1c and psychological outcomes for patients with diabetes mellitus.

Overall, MBIs have a moderate to high for improving HbA1c, and psychological outcomes (stress, anxiety and depression) in patients with diabetes mellitus compared to control groups. As we previously described, some *meta*-analysis researchers examined the effects of MBIs on HbA1c and psychological outcomes. Those who did include small number of primary studies (s = 4–9), which prohibited moderator analysis [21,22,27,28]. Therefore, our *meta*-analysis fills a gap in literature by including larger primary studies (s = 24) and conducting moderator analyses based on sources of information, methods, intervention, and participants' characteristics that suggest future research directions.

We found MBIs had a moderate effect on improving HbA1c level and psychological outcomes in patients with diabetes mellitus. These findings are not only statistically significant but also clinically meaningful: the observed effect sizes align with established thresholds for reducing diabetic complications and improving mental health symptoms. By tailoring mindfulness interventions to address the persistent psychosocial stressors of diabetes management, such as the burden of glucose monitoring and insulin administration, these statistical gains can translate into tangible, sustainable improvements in patient outcomes. To provide a deeper understanding of these outcomes, it is useful to view them through the lens of established clinical benchmarks. For HbA1c, even moderate reductions are associated with a substantial decrease in the risk of long-term microvascular complications, such as retinopathy, as evidenced by longitudinal data from the UKPDS [64]. For psychological outcomes, the effect sizes we observed meet the Minimal Clinically Important Difference (MCID) benchmarks—often defined as a change of approximately 5 points on standardized anxiety and depression scales—suggesting that patients experience real-world improvements in their daily functioning and quality of life [65].

Moving forward, the clinical utility of MBIs may be optimized by focusing on the neurophysiological mechanisms of mindfulness. By enhancing emotional regulation and prefrontal cortex function, mindfulness may help patients better navigate the "appetite-regulation" challenges inherent in dietary adherence, thereby bridging the gap between improved psychological health and sustained glycemic control [28,66].

These promising results support the usefulness of MBIs as alternative complementary treatments for patients with diabetes mellitus. They show solid evidence of their effectiveness in improving HbA1c and psychological outcomes (stress, anxiety, and depression). However, it is crucial to consider the mechanism of mindfulness practice in long-term effects of these interventions in different cultural settings.

5.2. Subgroup analyses

Conducting MBIs with funding was associated with a smaller effect size for HbA1c improvement compared to unfunded studies. One potential explanation is that funded researchers possess the resources to maintain higher methodological standards and recruit larger sample sizes, thereby increasing the precision of their estimates [32,67]. In our *meta*-analysis, funded research exhibited a mean sample size of 86.42 (SD = 37.35), compared to 63.33 (SD = 40.71) for unfunded studies. Because *meta*-analyses assign greater weight to studies with higher precision—which is fundamentally driven by sample size—funded projects likely benefit from greater statistical reliability [32]. This is further supported by the likelihood that funded studies more frequently incorporated a priori power analyses, a standard requirement for grant approval designed to ensure sufficient power to detect meaningful effect sizes.

Our findings indicate that individual-based MBIs yielded greater improvements in anxiety symptoms among patients with diabetes mellitus compared to group-based formats. The superior efficacy of individual sessions may stem from the capacity for personalized attention, enhanced confidentiality, and a customized pace, allowing for deeper engagement with sensitive emotional concerns [68,69]. However, the choice between these delivery formats involves distinct trade-offs. Individual sessions offer tailored interventions, whereas group formats provide unique social support and shared-learning benefits that may enhance adherence, though their efficacy can be contingent upon whether the curriculum is specifically adapted for the unique psychosocial challenges of diabetes management. Furthermore, the two formats differ significantly in their implementation; group-based MBIs typically require standardized instruction times and collective practice expectations, which contrast with the flexible, patient-centered pacing of individual sessions. Given these considerations, we recommend that healthcare professionals weigh the specific needs of the patients—such as the requirement for privacy versus the potential benefit of peer support—when selecting an MBIs format, while ensuring that group-based programs are carefully tailored to the diabetes population to maximize their therapeutic impact.

The application of the intention-to-treat methodology for analysis significantly enhanced HbA1c levels, stress, and depression outcomes compared to the absence of this strategy in individuals with diabetes mellitus. Intention-to-treat analysis is a conservative approach for evaluating intervention effects, as it includes all participants based on their initial group assignments [70–72]. Consequently, intention-to-treat analysis encompasses all individuals, even those who withdrew and exhibited no improvement. An intention-to-treat analysis delivers an accurate evaluation of the intervention's efficacy concerning adherence within the trial [70,72].

Moreover, using concealed allocation techniques had a less effect to improve HbA1c and depression in patients with diabetes mellitus than without technique. Concealed allocation refers to the process of hiding the random sequence of treatment assignments for researchers and participants in the original randomized controlled trial (RCTs). This technique is critical to prevent selection bias, ensuring that the characteristics of participants assigned to different treatment groups are truly random and not influenced by researchers' decisions [73,74]. So, this is vital for determining the reliability of the included trials' findings and their contribution to the overall evidence.

Interestingly, attrition rate is considered a moderator factor affecting the effectiveness of MBIs in patients with diabetes mellitus. We found that the attrition rate increased, the effects of MBIs was reduced, indicating an increase in HbA1c level. Since a higher attrition rate also resulted in a smaller number of participants in the analysis, the precision of the effect size is reduced [75,76]. Thus, we would recommend future researchers account for attrition during the recruitment process.

Ultimately, employing a priori analysis for participant recruitment had a lesser impact on ameliorating depression symptoms compared to the techniques that were not utilized. Researchers conducting interventions with a priori analysis has sufficient power to identify the impacts of MBIs [77]. Performing a power analysis before recruitment and factoring in attrition when determining sample size is crucial for ensuring that researchers attract and maintain a sufficient number of participants, which has significant ramifications for subsequent researchers [78]. Therefore, we recommend researchers to use a priori analysis before recruiting participants for accurate findings.

Although we hypothesized a dose–response relationship, *meta*-regression analysis of total intervention hours as a moderator yielded non-significant findings. This absence of a linear correlation suggests that for mindfulness-based interventions (MBIs) in diabetes management, cumulative instructional time—the traditional 'dose'—is not the sole determinant of therapeutic efficacy. Instead, our findings align with evidence indicating that outcomes are more heavily influenced by the quality of mindfulness practice, the degree of patient engagement, and the personalization of the curriculum to diabetes-specific stressors [61]. It is likely that MBIs operate under a 'threshold effect,' where a minimum baseline of training is required to achieve clinical gains, but additional hours beyond this threshold provide diminishing marginal returns [79,80]. Consequently, clinical consistency and the daily integration of mindfulness techniques appear to be more vital for improving HbA1c and psychological symptoms than the quantity of formal face-to-face sessions [28,81].

5.3. Implications and recommendations

As the findings above suggest, we recommend promoting the use of MBIs as an individual format for patients with diabetes mellitus as an alternative and complementary treatment for improving HbA1c and psychological outcomes. Mindfulness intervention may help reduce the severity of stress, anxiety, and depressive symptoms and prevent chronic mental health problems in the future. Healthcare policy makers might encourage clinicians to use MBIs as adjunct treatment for self-reported stress, anxiety, and depression in patients with diabetes mellitus. In situations where traditional physical/psychiatric care is not available or feasible, MBIs can be used as an alternative to help decrease the level of HbA1c and psychological outcomes and perhaps reduce the need for more expensive health care costs down the road.

For research implications, we recommend that future researchers consider examining the effects of MBIs on HbA1c and psychological outcomes in a longitudinal study. Additionally, researchers should conduct trials using strong methodological strategies, such as intention-to-treat analysis, concealed allocation, and a priori analysis, and account for attrition rates.

5.4. Strengths and limitations

Our research involved a comprehensive systematic review and *meta*-analysis examining the effects of mindfulness-based interventions on HbA1c levels and psychological outcomes specifically in individuals with diabetes mellitus. We also investigated the sources of heterogeneity in the findings. However, this *meta*-analysis has certain limitations. First, we only included research studies written in English, which may introduce language bias. Future research should consider incorporating studies in other languages to minimize this bias. Second, only five primary researchers assessed the effects of mindfulness-based interventions for more than three months after the intervention. Future studies could explore the long-term impacts of these interventions on HbA1c levels and psychological outcomes. Third, the inconsistent reporting of participant sociodemographic data—such as race, ethnicity, gender, and education level—across the included primary studies represents a significant limitation of this *meta*-analysis. The absence of such granular data precluded a robust moderator analysis, leaving it unclear whether MBI efficacy remains consistent across diverse populations or if benefits are skewed toward specific subgroups. Addressing these demographic gaps is essential for establishing the generalizability and equity of mindfulness interventions; therefore, future research must strictly adhere to standardized reporting guidelines, such as the CONSORT-Equity extension, to ensure transparent documentation of participant characteristics. Lastly, most primary researchers did not report any negative effects associated with mindfulness techniques. Thus, further research is necessary to investigate any potential adverse effects of mindfulness practices.

6. Conclusions

Mindfulness-based interventions (MBIs) may improve HbA1c levels and psychological outcomes, particularly in reducing stress, anxiety, and depression among individuals with diabetes mellitus. Additionally, we found that factors such as the format of MBIs, sources of funding, intention-to-treat methodology, allocation concealment, attrition rate, and planned analyses influenced the effect size of these interventions. However, it's notable that primary researchers did not disclose any adverse effects related to mindfulness practices. As a result, further investigation into the safety and potential negative effects of MBIs is warranted. More research is also needed to evaluate the long-term effectiveness of mindfulness therapies in treating depression, with an emphasis on employing stronger methodological rigor.

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CRediT authorship contribution statement

Chuntana Reangsing: Writing – review & editing, Writing – original draft, Software, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Sasinun Punsuwun:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Data curation, Conceptualization. **Khanittha Pitchalard:** Writing – review & editing, Writing – original draft, Validation, Investigation, Data curation. **Sarah Oerther:** Writing – review & editing, Writing – original draft, Validation, Supervision, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] Guan H, et al. Advances in secondary prevention mechanisms of macrovascular complications in type 2 diabetes mellitus patients: a comprehensive review. Eur J Med Res 2024;29(1):152.
[2] Zakir M, et al. Cardiovascular complications of diabetes: from microvascular to macrovascular pathways. Cureus 2023;15(9):e45835.
[3] Chen J, Yin D, Dou K. Intensified glycemic control by HbA1c for patients with coronary heart disease and Type 2 diabetes: a review of findings and conclusions. Cardiovasc Diabetol 2023;22(1):146.
[4] Ying DGC, et al. Association between intensive glycemic control and mortality in elderly diabetic patients in the primary care: a retrospective cohort study. Prim Care Diabetes 2020;14(5):476–81.
[5] Soff S, et al. Association of glycemic control with Long COVID in patients with type 2 diabetes: findings from the National COVID Cohort Collaborative (N3C). BMJ Open Diabetes Res Care 2025;13(1):e004536.
[6] Boye KS, et al. The association between sustained HbA1c control and long-term complications among individuals with type 2 diabetes: a retrospective study. Adv Ther 2022;39(5):2208–21.
[7] Wojujutari AK, Idemudia ES, Ugwu LE. Psychological resilience mediates the relationship between diabetes distress and depression among persons with diabetes in a multi-group analysis. Sci Rep 2024;14(1):6510.
[8] Fanelli G, et al. The interface of depression and diabetes: Treatment considerations. Transl Psychiatry 2025;15(1):22.
[9] McCoy MA, Theeke LA. A systematic review of the relationships among psychosocial factors and coping in adults with type 2 diabetes mellitus. International Journal of Nursing Sciences 2019;6(4):468–77.
[10] Kalra S, Jena BN, Yeravdekar R. Emotional and psychological needs of people with diabetes. Indian Journal of Endocrinology and Metabolism 2018;22(5):696–704.
[11] Weerasinghe SA, et al. Psychosocial interventions for diabetes distress in culturally diverse populations: a systematic scoping review. BMJ Open 2025;15(10).
[12] Widiastuti L, et al. Culturally tailored education interventions to enhance diabetes self-management: a systematic review of randomised controlled trials. J Multidiscip Healthc 2026;19.
[13] Yakubu TI, et al. Impact of digitally enabled peer support interventions on diabetes distress and depressive symptoms in people living with type 1 diabetes: a systematic review. Curr Diab Rep 2025;25(1).
[14] Press HS, Nicklas L. How effective are CBT and CBT- based interventions in Type 1 and Type 2 diabetes? An umbrella review Diabetic Medicine 2026;43:e70271.
[15] Ernawati U, Wihastuti TA, Utami YW. Effectiveness of diabetes self-management education (DSME) in type 2 diabetes mellitus (T2DM) patients: Systematic literature review. Journal of Public Health Research 2021;10(2240):1–5.
[16] Jerawatana R, et al. The effects of diabetes self-management education and support program in Thailand: a systematic review and meta-analysis. The Journal of the Medical Association of Thailand 2021;104(6):1050–8.
[17] Liu Y, et al. Community-based peer support for diabetes management: 24-month changes relative to comparison communities. Diabetes Care 2025;48:807–15.
[18] Kumoji Ayamsegna HN, et al. Effectiveness of peer-based interventions among adolescents with type 1 diabetes: a systematic review. J Pediatr Nurs 2026;86: 90–104.
[19] Varshini RJ, et al. Barriers and facilitators of implementing cognitive behavioral therapy: a systematic review based on the consolidated framework for implementation. Scientifica 2025.

[20] Tharakan, A., et al., Accessibility of diabetes education in the United States: barriers, policy implications, and the road ahead. Health Affairs Scholar, 2024. **2** (8): p. qxae097.
[21] Ee CC, et al. Mindfulness-based interventions for adults with type 2 diabetes mellitus: a systematic review and meta-analysis. Integrative Medicine Research 2025;14(2).
[22] Paez-Marquez CS, Higuera-Dagovett E, Rojas-Valencia JT. Effect of mindfulness-based intervention in patients with diabetes: a meta-analytic review. Psychol Health Med 2025:1–24.
[23] Jiang S, et al. Global trends and developments in mindfulness interventions for diabetes: a bibliometric study. Diabetol Metab Syndr 2024;16(1).
[24] Yusuf, A., et al., Integrating mindfulness and physical activity for diabetes prevention and management: Systematic review. Retos: fue fundada por la Federación Española de Docentes de Educación Física, 2024(59): p. 767-775.
[25] Kabat-Zinn J. Mindfulness-based interventions in context: past, present, and future. Clin Psychol Sci Pract 2003;10(2):144–55.
[26] Schuman-Olivier Z, et al. Mindfulness and behavior change. Harv Rev Psychiatry 2020;28(6):371–94.
[27] Ngan HY, Chong YY, Chien WT. Effects of mindfulness- and acceptance-based interventions on diabetes distress and glycaemic level in people with type 2 diabetes: Systematic review and meta-analysis. Diabet Med 2021;38(4).
[28] Hamasaki H. The effects of mindfulness on glycemic control in people with diabetes: an overview of systematic reviews and meta-analyses. Medicines 2023;10 (53).
[29] Moher D, et al. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. BMJ 2009;339:b2535.
[30] Page MJ, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71.
[31] Frandsen TF, et al. Using the full PICO model as a search tool for systematic reviews resulted in lower recall for some PICO elements. J Clin Epidemiol 2020; 127:69–75.
[32] Borenstein M, et al. Introduction to meta-analysis. 2nd ed. Oxford: Wiley & Sons Ltd.; 2021.
[33] van Aert R, Jackson D. A new justification of the HartungKnapp method for randomeffects metaanalysis based on weighted least squares regression. Res Synth Methods 2019.
[34] Jackson D, et al. The Hartung-Knapp modification for random-effects meta-analysis: a useful refinement but are there any residual concerns? Stat Med 2017; 36:3923–34.
[35] Higgins JPT, Green S. Cochrane handbook for systematic review of interventions version 5.1.0. The Cochrane Collaboration 2011.
[36] Meader N, et al. A checklist designed to aid consistency and reproducibility of GRADE assessments: development and pilot validation. Syst Rev 2014;3(82).
[37] Shao SC, et al. Using Grading of Recommendations Assessment, Development, and Evaluation (GRADE) to rate the certainty of evidence of study outcomes from systematic reviews: a quick tutorial. Dermatology Sinica 2023;41:3–7.
[38] Hamidi F, et al. The effectiveness of mindfulness-based training on anxiety in pregnant women with gestational diabetes. Family Medicine & Primary Care Review 2020;22(4):279–83.
[39] Kian AA, et al. The impact of mindfulness-based stress reduction on emotional wellbeing and glycemic control of patients with type 2 diabetes mellitus. Journal of Diabetes Research 2018;2018.
[40] Ravari OM, Mousavi SZ, Babak A. Evaluation of the effects of 12 weeks mindfulness-based stress reduction on glycemic control and mental health indices in women with diabetes mellitus type 2. Advanced Biomedical Research 2020;9:61.
[41] Sohrabi F, et al. Managing type 2 diabetes and depression with Mindfulness-integrated Cognitive Behavior Therapy (MiCBT). Discov Psychol 2022;2(1).
[42] Zarifsanaiey N, et al. The effects of mindfulness training on the level of happiness and blood sugar in diabetes patients. J Diabetes Metab Disord 2020;19(1):311–7.
[43] Chen Q, et al. The effects of mindfulness-based stress reduction therapy combined with intensive education on the effectiveness of the care and the awareness rate in patients with arthritis and diabetes. Am J Transl Res 2021;13(4):3190–7.
[44] Guo J, et al. Effectiveness of a nurse-led mindfulness stress-reduction intervention on diabetes distress, diabetes self-management, and HbA1c levels among people with type 2 diabetes: a pilot randomized controlled trial. Res Nurs Health 2022;45 (1):46–58.
[45] Wang H, et al. A blended mindfulness-based stress reduction program to improve diabetes self-management among people with type 2 diabetes mellitus: a mediation effect analysis. Ann Behav Med 2025;59(1).
[46] Xiao LH. Blood sugar reduction of type 2 diabetic patients through a mindfulness intervention program. Neuro Quantology 2018;16(1):57–62.
[47] Nyklicek I, et al. Does mindfulness-based cognitive therapy benefit all people with diabetes and comorbid emotional complaints equally? Moderators in the DiaMind trial. J Psychosom Res 2016;91:40–7.
[48] Tovote KA, et al. Individual mindfulness-based cognitive therapy and cognitive behavior therapy for treating depressive symptoms in patients with diabetes: results of a randomized controlled trial. Diabetes Care 2014;37:2427–34.
[49] van Son J, et al. Mindfulness-based cognitive therapy for people with diabetes and emotional problems: long-term follow-up findings from the DiaMind randomized controlled trial. J Psychosom Res 2014;77(1):81–4.
[50] DiNardo MM, et al. Effects of an integrated mindfulness intervention for veterans with diabetes distress: a randomized controlled trial. BMJ Open Diabetes Res Care 2022;10(2).
[51] Ellis DA, et al. Efficacy of mindfulness-based stress reduction in emerging adults with poorly controlled, type 1 diabetes: a pilot randomized controlled trial. Pediatr Diabetes 2019;20(2):226–34.
[52] Hartmann M, et al. Sustained effects of a mindfulness-based stress-reduction intervention in type 2 diabetic patients: Design and first results of a randomized controlled trial (the Heidelberger Diabetes and Stress-study). Diabetes Care 2012; 35(5):945–7.
[53] Kopf S, et al. Effects of stress reduction on cardiovascular risk factors in type 2 diabetes patients with early kidney disease – results of a randomized controlled trial (HEIDIS). Exp Clin Endocrinol Diabetes 2014;122:341–9.
[54] Patel J, Prakasam A. Effectiveness of mindfulness on depression, anxiety and stress among patients with type 2 diabetes. African Journal of Biological Sciences (South Africa) 2024;6(5):4522–38.
[55] Shukla R, et al. Mindfulness meditation as adjunctive therapy to improve the glycemic care and quality of life in patients with type 1 diabetes. Medical Sciences (Basel, Switzerland) 2021;9(2).
[56] Friis AM, et al. Kindness matters: a randomized controlled trial of a mindful self-compassion intervention improves depression, distress, and HbA1c among patients with diabetes. Diabetes Care 2016;39:1963–71.
[57] Hosseini SS, et al. Effectiveness of mindfulness-based therapy on resilience, psychological well-being, and blood sugar levels in patients with type II diabetes. Razavi. International Journal of Medicine 2021;9(2).
[58] Jung HY, Lee H, Park J. Comparison of the effects of Korean mindfulness-based stress reduction, walking, and patient education in diabetes mellitus. Nurs Health Sci 2015;17(4):516–25.
[59] Nathan HJ, et al. Randomized trial of the effect of mindfulness-based stress reduction on pain-related disability, pain intensity, health-related quality of life, and A1C in patients with painful diabetic peripheral neuropathy. Clin Diabetes 2017:294–304.
[60] Pearson S, et al. Effects of mindfulness on psychological distress and HbA1c in people with diabetes. Mindfulness 2018;9(5):1615–26.
[61] Sukchaisong N, et al. Effectiveness of the mindfulness-based diabetes self- and family management support program among adults with uncontrolled diabetes: a randomized controlled trial. Pacific Rim International Journal of Nursing Research 2022;26(3):517–32.
[62] Schünemann, H.J., et al., Completing 'Summary of findings' tables and grading the certainty of the evidence in Cochrane Handbook for Systematic Reviews of Interventions version 6.5. Cochrane, J.P.T. Higgins, et al., Editors. 2024.
[63] Brennan SE, Johnston RV. Research note: Interpreting findings of a systematic review using GRADE methods. J Physiother 2023;69:198–202.
[64] Stratton IM, et al. Association of glycaemia with macrovascular and microvascular complications of type 2 diabetes (UKPDS 35): prospective observational study. BMJ 2000;321(7258):405.
[65] Cuijpers P, et al. Psychological treatment of depression: a meta-analytic database of randomized studies. BMC Psychiatry 2008;8:36.
[66] Qona'ah A, et al. Impact of mindfulness-based stress reduction on diabetes mellitus psychological well-being and glycemic control: a systematic Review. Gaceta Medica de Caracas 2024;132:s294–304.
[67] Hansen, C., et al., Financial conflicts of interest in systematic reviews: Associations with results, conclusions, and methodological quality Cochrane Database of Systematic Reviews, 2019(8): p. MR000047.
[68] Guo J, et al. Factors influencing the effect of mindfulness-based interventions on diabetes distress: a meta-analysis. BMJ Open Diabetes Res Care 2019;7(1): e000757.
[69] Cuijpers, P., A. van Straten, and L. Warmerdam, Are individual and group treatments equally effective in the treatment of depression in adults? A meta-analysis. 2008, European Journal of Psychiatry: Spain. p. 38-51.
[70] McCoy EC. Understanding the intention-to-treat principle in randomized controlled trials. Western Journal of Emergency Medicine 2017;18(6):1075–8.
[71] Gupta SK. Intention-to-treat concept: a review. Perspect Clin Res 2011;2(3): 109–12.
[72] Armijo-Olivo S, et al. Intention-to-treat analysis in clinical research: basic concepts for clinicians. Am J Phys Med Rehabil 2024;103(9):845–57.
[73] Zeraatkar D, et al. Impact of allocation concealment and blinding in trials addressing treatments for COVID-19: a methods study. Am J Epidemiol 2023;192 (10):1678–87.
[74] Schulz KF, Grimes DA. Allocation concealment in randomised trials: defending against deciphering. Lancet 2002;359(9306):614–8.
[75] Gustavson K, et al. Attrition and generalizability in longitudinal studies: findings from a 15-year population-based study and a Monte Carlo simulation study. BMC Public Health 2012;12(1):918.
[76] Bevens W, et al. Systematic review and meta analysis of differential attrition between active and control arms in randomized controlled trials of lifestyle interventions in chronic disease. BMC Med Res Method 2021;21(1):122.
[77] Beck TW. The importance of a priori sample size estimation in strength and conditioning research. J Strength Cond Res 2013;27(8).
[78] Abt G, et al. Power, precision, and sample size estimation in sport and exercise science research. J Sports Sci 2020;38(17):1933–5.
[79] Khoury B, et al. Mindfulness-based therapy: a comprehensive meta-analysis. Clin Psychol Rev 2013;33(6):763–71.
[80] Parsons CE, et al. Home practice in Mindfulness-based Cognitive Therapy and Mindfulness-based stress Reduction: a systematic review and meta-analysis of participants' mindfulness practice and its association with outcomes. Behav Res Ther 2017;95:29–41.
[81] Creswell, J.D., Mindfulness interventions. Annual Review of Psychology, 2017. **68** (Volume 68, 2017): p. 491-516.